

One-Bit DOA Estimation via $\ell_{2,0}$ -Regularized Sparse Recovery

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Abstract—One-bit quantization can lower power consumption and data transmission cost, but it introduces severe nonlinearity and amplitude loss, which complicates direction-of-arrival (DOA) estimation. This paper proposes a one-bit DOA estimation method based on maximum likelihood modeling and row-sparse matrix recovery with $\ell_{2,0}$ -norm regularization. By deriving the distribution of one-bit observations, the DOA estimation problem is formulated as a sparse recovery task. The iterative hard thresholding algorithm with an adaptively selected threshold based on the normalized median absolute deviation is developed. Simulation results demonstrate that the proposed method outperforms the existing approaches.

Index Terms—DOA estimation, one-bit quantization, maximum likelihood (ML), row-sparse matrix recovery

I. INTRODUCTION

Direction-of-arrival (DOA) estimation is a fundamental problem in array signal processing, and has broad applications in radar, wireless communications, and sonar systems [1]. Recently, one-bit analog-to-digital converters (ADCs) have attracted increasing attention due to their extremely simple hardware architecture, and thus are suitable for resource-constrained platforms (e.g., lightweight unmanned aerial vehicles and autonomous systems) [2]. Nevertheless, achieving accurate DOA estimation from one-bit measurements remains a significant challenge. The main challenge stems from the fact that one-bit quantization retains only the signal sign, creating a highly nonlinear relationship between the quantized and unquantized signals. As a result, conventional DOA methods have performance degradation.

Subspace-based one-bit DOA estimation primarily relies on the arcsine law [3], which establishes a nonlinear mapping between the covariance matrices of quantized and unquantized signals. Then, various one-bit DOA estimation manners for uniform linear arrays (ULAs) and sparse linear array (SLAs) have been developed [4]–[7]. In addition, covariance matrix recovery with non-zero and time-varying quantization thresholds is devised [8]. However, these approaches require the prior knowledge of the source number. In practice, source number estimation under one-bit quantization is challenging [9]–[11].

Compared with the subspace-based approaches, compressed sensing (CS)-based DOA estimation methods that exploit the spatial sparsity of the source signal have better performance in scenarios with low signal-to-noise ratio (SNR), a limited snapshot number, and coherent sources [12], [13]. Representative algorithms include complex-valued binary iterative hard thresholding (CBIHT) and its improved variants [14], [15], atomic norm denoising methods [16], generalized sparse Bayesian learning [17], off-grid iterative reweighting techniques [18], and robust single-bit DOA estimation method [19]. In addition, robust sparse covariance fitting-based approaches have been developed to address one-bit DOA estimation under nonuniform noise conditions [20].

Despite the above approaches have made progress, there is a need to devise a method that can directly exploit the probability distribution of one-bit measurements and has the robustness in low-SNR and limited-snapshot scenarios. Therefore, this paper develops a one-bit likelihood model and formulate the DOA estimation problem by a grid-based row-sparse matrix recovery task. The resultant nonconvex problem is solved using the iterative hard-thresholding (IHT) procedure, where the row-selection threshold is adaptively updated using normalized median absolute deviation (NMAD) [21].

II. SIGNAL MODEL

P far-field narrowband source signals $s_p(t)$ with $p = 1, 2, \dots, P$ impinge on a ULA consisting of M antenna elements. The inter-element spacing of the ULA is d . The

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DOA of the p -th source denoted by θ_p is defined as the angle between the incident signal and the array broadside.

In the presence of additive noise, the unquantized baseband signal received by the m -th array element at the t -th snapshot is given by

$$y_m(t) = \sum_{p=1}^P e^{i2\pi(m-1)\frac{d \sin \theta_p}{\nu}} s_p(t) + v_m(t), \quad (1)$$

where ν denotes the signal wavelength, $i = \sqrt{-1}$ is the imaginary unit, and $v_m(t) \sim \mathcal{CN}(0, \sigma^2)$ is additive noise with σ^2 being the noise variance. We assume that the noise samples are statistically independent across antenna elements and snapshots. This assumption is common in one-bit likelihood modeling and are adopted to obtain a tractable formulation. Extensions to more general case are left for future work.

One-bit analog-to-digital converters act on the real and imaginary parts of the received signal separately. Consequently, the one-bit quantized output is expressed as

$$z_m(t) = \text{sgn}(\Re(y_m(t))) + i \text{sgn}(\Im(y_m(t))), \quad (2)$$

where $\Re(\cdot)$ and $\Im(\cdot)$ denote the real and imaginary parts of a complex-valued number, respectively. Then, the signals received by the M array elements at the t -th snapshot can be formulated as

$$\mathbf{z}(t) = \mathcal{Q}(\mathbf{y}(t)) = \mathcal{Q}(\mathbf{A}(\boldsymbol{\theta})\mathbf{s}(t) + \mathbf{v}(t)), \quad (3)$$

where $\mathcal{Q}(\cdot)$ is the one-bit quantization operator. In addition, the array steering vector corresponding to θ_p is defined as $\mathbf{a}(\theta_p) = [1 \ e^{i2\pi\frac{d \sin \theta_p}{\nu}} \ \dots \ e^{i2\pi(M-1)\frac{d \sin \theta_p}{\nu}}]^T$. Accordingly, the array manifold matrix is given by $\mathbf{A}(\boldsymbol{\theta}) = [\mathbf{a}(\theta_1) \ \mathbf{a}(\theta_2) \ \dots \ \mathbf{a}(\theta_P)]$, where $\boldsymbol{\theta} = [\theta_1, \theta_2, \dots, \theta_P]^T$ denotes the DOAs of P sources.

By stacking the received signals over T snapshots, we have

$$\mathbf{Z} = \mathcal{Q}(\mathbf{Y}) = \mathcal{Q}(\mathbf{A}(\boldsymbol{\theta})\mathbf{S} + \mathbf{V}). \quad (4)$$

The aim of one-bit DOA estimation is to estimate $\boldsymbol{\theta}$ based on $\mathbf{Z}$.

III. PROPOSED METHOD

A. One-bit Maximum Likelihood Estimation

Under the assumption of circularly symmetric complex Gaussian noise, the real and imaginary parts of the unquantized signal $y_m(t)$ are statistically independent. In particular, $\Re(y_m(t)) \sim \mathcal{N}(\Re(\mathbf{A}_m(\boldsymbol{\theta})\mathbf{s}(t)), \sigma^2/2)$ and $\Im(y_m(t)) \sim \mathcal{N}(\Im(\mathbf{A}_m(\boldsymbol{\theta})\mathbf{s}(t)), \sigma^2/2)$, where $\mathbf{A}_m(\boldsymbol{\theta})$ denotes the m -th row of $\mathbf{A}(\boldsymbol{\theta})$. Therefore, the joint probability mass function of $z_m(t)$ is expressed as

$$\Pr(z_m(t)) = \Pr(\Re(z_m(t))) \Pr(\Im(z_m(t))), \quad (5)$$

where

$$\Pr(\Re(z_m(t))) = \Phi\left(\frac{\sqrt{2} \Re(z_m(t)) \Re(\mathbf{A}_m(\boldsymbol{\theta})\mathbf{s}(t))}{\sigma}\right), \quad (6)$$

$$\Pr(\Im(z_m(t))) = \Phi\left(\frac{\sqrt{2} \Im(z_m(t)) \Im(\mathbf{A}_m(\boldsymbol{\theta})\mathbf{s}(t))}{\sigma}\right), \quad (7)$$

and $\Phi(\cdot)$ denotes the cumulative distribution function of a standard normal random variable. Herein, (6) and (7) compactly characterize the probabilities associated with the events $\{\Re(z_m(t)) = \pm 1 \text{ and } \Im(z_m(t)) = \pm 1\}$, respectively.

Considering M array elements and T snapshots, the likelihood function of $\mathbf{Z}$ is given by

$$\begin{aligned} \mathcal{L}(\boldsymbol{\theta}, \mathbf{S}, \sigma) = & \sum_{t=1}^T \sum_{m=1}^M \ln \left(\Phi\left(\frac{\sqrt{2} \Re(z_m(t)) \Re(\mathbf{A}_m(\boldsymbol{\theta})\mathbf{s}(t))}{\sigma}\right) \right) \\ & + \sum_{t=1}^T \sum_{m=1}^M \ln \left(\Phi\left(\frac{\sqrt{2} \Im(z_m(t)) \Im(\mathbf{A}_m(\boldsymbol{\theta})\mathbf{s}(t))}{\sigma}\right) \right). \end{aligned} \quad (8)$$

Based on the derived log-likelihood function, the one-bit maximum likelihood estimation (MLE) of $\boldsymbol{\theta}$ can be formulated as:

$$\hat{\boldsymbol{\theta}} = \arg \max_{\boldsymbol{\theta}, \mathbf{S}, \sigma} \mathcal{L}(\boldsymbol{\theta}, \mathbf{S}, \sigma). \quad (9)$$

However, this optimization problem is nonconvex and non-linear, making it difficult to solve efficiently. In particular, the log-likelihood function $\mathcal{L}(\boldsymbol{\theta}, \mathbf{S}, \sigma)$ is multimodal with respect to $\boldsymbol{\theta}$, which makes direct optimization over the continuous angle space computationally difficult and highly sensitive to initialization.

B. Sparse Matrix Recovery

Motivated by the CS framework, (9) can be reformulated as a sparse matrix recovery problem by exploiting the inherent spatial sparsity of the signal matrix. To this end, the entire angular domain is discretized into a dense grid $\boldsymbol{\Theta} = \{\tilde{\theta}_1, \tilde{\theta}_2, \dots, \tilde{\theta}_G\}$, where $G \gg P$. Let $\mathbf{A}(\boldsymbol{\Theta}) \in \mathbb{C}^{M \times G}$ be the corresponding overcomplete array manifold matrix, whose g -th column $\mathbf{a}(\tilde{\theta}_g)$ represents the steering vector associated with direction $\tilde{\theta}_g$. Furthermore, we define a row-sparse matrix $\mathbf{X} = [\mathbf{x}(1), \dots, \mathbf{x}(T)] \in \mathbb{C}^{G \times T}$, whose rows satisfy the following equality

$$\mathbf{X}_g = \frac{\sqrt{2}}{\sigma} \mathbf{S}_p, \text{ if } \tilde{\theta}_g = \theta_p. \quad (10)$$

It is worth mentioning that $\mathbf{S} \in \mathbb{C}^{P \times T}$ is embedded in $\mathbf{X}$, while $\boldsymbol{\theta}$ is encoded in the row support of $\mathbf{X}$. As a result, the joint estimation of $(\boldsymbol{\theta}, \mathbf{S}, \sigma)$ in the original MLE formulation is equivalently transformed into the recovery of $\mathbf{X}$, leading to

$$\min_{\mathbf{X}} \tilde{\mathcal{L}}(\mathbf{X}) + \lambda \|\mathbf{X}\|_{2,0}, \quad (11)$$

where $\lambda > 0$ is a regularization parameter to control the sparsity level and $\|\mathbf{X}\|_{2,0} = \|[\|\mathbf{x}_{1,:}\|_2, \dots, \|\mathbf{x}_{M,:}\|_2]^T\|_0$ is $\ell_{2,0}$ -norm, corresponding to the number of nonzero rows in $\mathbf{X}$. Then, the negative log-likelihood function is expressed as

$$\begin{aligned} \tilde{\mathcal{L}}(\mathbf{X}) = & \sum_{t=1}^T \sum_{m=1}^M -\ln(\Phi(\Re(z_m(t)) \Re(\mathbf{A}_m(\boldsymbol{\Theta})\mathbf{x}(t)))) \\ & + \sum_{t=1}^T \sum_{m=1}^M -\ln(\Phi(\Im(z_m(t)) \Im(\mathbf{A}_m(\boldsymbol{\Theta})\mathbf{x}(t)))). \end{aligned} \quad (12)$$

Once $\mathbf{X}$ is recovered, the DOA estimates can be obtained by identifying the indices of the nonzero rows, and mapping them back to the grid angles in Θ .

C. Iterative Hard Thresholding Based on Normalized Median Absolute Deviation

It is evident that (11) is a non-smooth and non-convex unconstrained problem. To address this issue, we adopt an iterative hard-thresholding (IHT) framework that alternates between a gradient-based update and a row-wise hard-thresholding operation. Specifically, at the r -th iteration, the IHT procedure consists of the following two steps.

1) Gradient descent step: The likelihood term is optimized via a gradient descent update, given by

$$\mathbf{X}^{(r)} = \mathbf{X}^{(r-1)} - \eta \nabla \tilde{\mathcal{L}}(\mathbf{X}^{(r-1)}), \quad (13)$$

where η denotes the step size and $\nabla \tilde{\mathcal{L}}(\mathbf{X}^{(r-1)})$ is the gradient of the negative log-likelihood function $\tilde{\mathcal{L}}(\mathbf{X})$ with respect to $\mathbf{X}$. The detailed derivation of the gradient expression can be found in [22].

2) Hard-thresholding step: After the gradient descent update, a row-wise hard-thresholding operation is applied to promote sparsity. Specifically, the g -th row of $\mathbf{X}^{(r)}$ is updated as [23]

$$\mathbf{X}_g^{(r)} = \begin{cases} \mathbf{X}_g^{(r)}, & \|\mathbf{X}_g^{(r)}\|_2 > \sqrt{\lambda}, \\ \mathbf{0}, & \text{otherwise,} \end{cases} \quad (14)$$

where $\|\cdot\|_2$ denotes the ℓ_2 -norm.

A critical issue in the IHT framework lies in the selection of λ . Unlike conventional approaches that rely on the prior knowledge of the sparsity level or noise variance, this paper investigates the threshold selection problem from a novel perspective based on spatial spectrum analysis.

As discussed in the previous section, each source directions associated with a nonzero row of the row-sparse matrix. Accordingly, a spatial spectrum can be constructed by computing the row-wise ℓ_2 -norm of the recovered matrix, given by

$$\gamma^{(r)} = \left[\|\mathbf{X}_1^{(r)}\|_2, \|\mathbf{X}_2^{(r)}\|_2, \dots, \|\mathbf{X}_G^{(r)}\|_2 \right]^T. \quad (15)$$

As illustrated in Fig. 1, only a few spectral components have significantly larger magnitudes, representing the true source directions, while the remaining components correspond primarily to background noise or spurious responses. By applying an appropriate threshold, indicated by the red dashed line in Fig. 1, these dominant rows can be reliably identified, enabling accurate detection of DOAs. The simulation setting used to generate Fig. 1 is described in Section IV. In the proposed framework, this threshold is determined by λ . Inspired by robust statistical techniques in [24], [25], the normalized median absolute deviation (NMAD) is employed to adaptively determine the threshold at each iteration. The standard deviation estimator of $\gamma^{(r)}$ is computed as

$$\rho^{(r)} = 1.4826 \times \text{Med} \left(\left| \gamma^{(r)} - \text{Med}(\gamma^{(r)}) \right| \right), \quad (16)$$

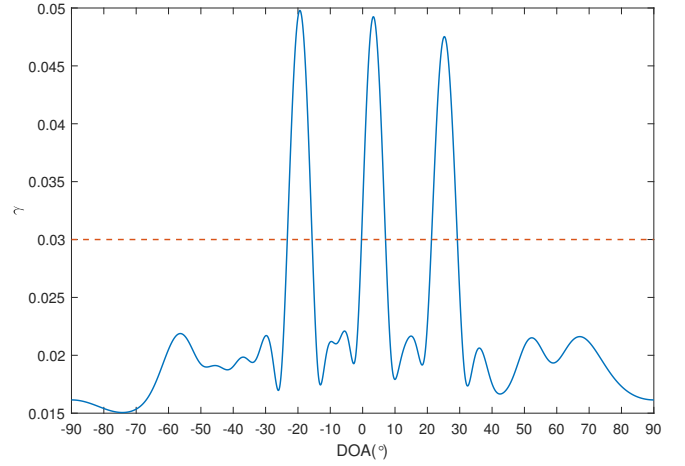


Fig. 1. Spatial spectrum diagram of row sparse matrix

where $\text{Med}(\cdot)$ denotes the median operator and the factor 1.4826 is the standard Gaussian consistency constant. Then, $\gamma^{(r)}$ is sorted in descending order to obtain

$$\gamma_{\text{sort}}^{(r)} = [\gamma_1^{(r)}, \gamma_2^{(r)}, \dots, \gamma_G^{(r)}], \gamma_1^{(r)} \geq \gamma_2^{(r)} \geq \dots \geq \gamma_G^{(r)}. \quad (17)$$

To avoid the influence of strong source components, the largest $D = \lceil 0.01G \rceil$ elements are removed, yielding the remaining spectrum

$$\gamma_{\text{remain}}^{(r)} = [\gamma_{D+1}^{(r)}, \gamma_{D+2}^{(r)}, \dots, \gamma_G^{(r)}], \quad (18)$$

where $\lceil \cdot \rceil$ denotes the ceiling operator. Then, the mean of the remaining spectrum is computed as

$$\mu^{(r)} = \frac{1}{G-D} \sum_{g=D+1}^G \gamma_g^{(r)}. \quad (19)$$

Finally, the threshold parameter is updated as

$$\lambda^{(r)} = (\mu^{(r)} + \zeta \rho^{(r)})^2, \quad (20)$$

where ζ denotes a standard deviation shrinkage factor. Although the shrinkage factor is introduced, simulation results in the subsequent section demonstrate that the proposed method is robust to its value, and thus ζ can be chosen in a straightforward manner.

In terms of computational complexity, the gradient descent step is on the order of $\mathcal{O}(MGT)$. In addition, the row-wise hard-thresholding operation and the construction of the spatial spectrum require $\mathcal{O}(GT)$, while sorting incurs a complexity of $\mathcal{O}(G \log G)$. Therefore, the overall computational complexity of the proposed algorithm is $\mathcal{O}(R(MGT + G \log G))$, where R is the number of iterations.

IV. SIMULATION

Two simulations are conducted to evaluate the proposed method, focusing on the impact of the standard deviation shrinkage operator and the estimation performance under different SNRs, respectively. The Cramér–Rao bound (CRB) for arbitrary unbiased estimators [26] is included for comparison.

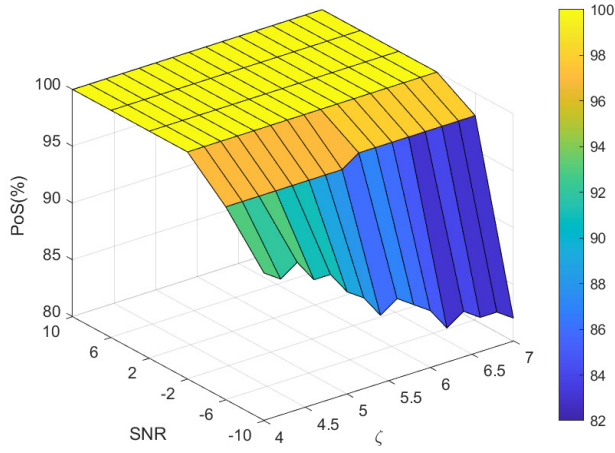


Fig. 2. Success probability versus SNR

The first simulation evaluates the robustness of the proposed algorithm to the choice of the standard deviation shrinkage operator under varying SNR conditions. Three far-field narrow-band signals $s_1(t) = e^{j\frac{\pi}{15}t}$, $s_2(t) = e^{j\frac{\pi}{3}t}$, and $s_3(t) = e^{j\frac{8\pi}{15}t}$ impinge on the ULA from directions -33° , 2° , and 23° , respectively. In the second simulation, the DOA estimation accuracy is quantified by root mean square error (RMSE) and the competing methods include one-bit MUSIC [6], OGIR [18], AND [16], and Gr-SBL [17]. The source signals are given by $s_1(t) = e^{j\frac{2\pi}{50}t}$, $s_2(t) = e^{j\frac{\pi}{5}t}$ and $s_3(t) = e^{j\frac{8\pi}{25}t}$ arriving from directions -19° , 3° , and 25° , respectively. The snapshot and antennas number are fixed at $T = 60$ and $M = 18$, respectively. The parameter settings of the proposed algorithm are summarized as follows. The number of angular grids is set to $G = 1801$, the step size is $\eta = 10^{-4}$, and the maximum number of iterations is $R = 500$. The initial values are chosen as $\mathbf{X}^0 = \mathbf{A}(\Theta)^H \mathbf{Z}$, $\lambda^0 = 1000$, and Monte Carlo trials are 100.

Fig. 2 illustrates the success probability of one-bit DOA estimation under different standard deviation shrinkage operators and SNR levels. It is observed that the success probability increases monotonically with SNR. In particular, when $\text{SNR} \geq -2$ dB, the proposed algorithm achieves a 100% success probability over the entire SNR range. Moreover, even for other choices of the shrinkage operator, the success probability remains above 90% over the tested SNR range. These results demonstrate the robustness of the proposed NMAD-based thresholding strategy. Based on this observation, the shrinkage operator is fixed to $\zeta = 4.4$ in the following simulation.

Fig. 3 compares the estimation performance of different algorithms with varying SNR. It is seen that the RMSE of all methods decreases as SNR increases, which is consistent with theoretical expectations. In addition, the proposed method consistently outperforms the competing methods across the entire SNR range, with a particularly significant advantage in the low-SNR regime. Furthermore, when the SNR exceeds 2 dB, the RMSE of the suggested method approaches

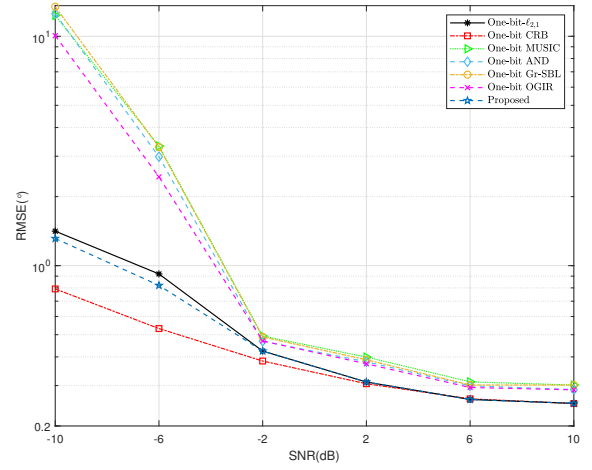


Fig. 3. RMSE versus SNR

the CRB, indicating that it achieves near-optimal estimation performance. These results confirm that the proposed method provides superior estimation accuracy over the existing algorithms, while reduces the reliance on manually tuned sparsity-related parameter.

V. CONCLUSION

To address the DOA estimation task with one-bit quantization, this paper proposes a novel method that integrates a likelihood-based formulation with sparse matrix recovery. The regularization parameter is adaptively updated using NMAD. Simulation results demonstrate that the standard deviation shrinkage operator exhibits weak sensitivity to key system parameter in the tested scenarios and the proposed method achieves superior DOA estimation accuracy over several representative methods, particularly under low SNR.

REFERENCES

- [1] P. Stoica and R. Moses, *Spectral Analysis of Signals*. Upper Saddle River, New Jersey: Prentice Hall, 2005.
- [2] M. Chen, Y.-H. Xiao, Q. Li, L. Feng, M. Rihan, and N. Guo, "One-bit source number detection via neural network," in *Int. Conf. Inf. Commun. Signal Process.*, (Shenzhen, China), pp. 91–95, Nov. 2022.
- [3] J. H. Van Vleck and D. Middleton, "The spectrum of clipped noise," *Proc. IEEE*, vol. 54, pp. 2–19, Jan. 1966.
- [4] O. Bar-Shalom and A. J. Weiss, "DOA estimation using one-bit quantized measurements," *IEEE Trans. Aerosp. Electron. Syst.*, vol. 38, pp. 868–884, Jul. 2002.
- [5] C.-L. Liu and P. P. Vaidyanathan, "One-bit sparse array DOA estimation," in *Proc. IEEE Int. Conf. Acoust., Speech, Signal Process.*, (New Orleans, LA, USA), pp. 3126–3130, Mar. 2017.
- [6] X. Huang and B. Liao, "One-bit MUSIC," *IEEE Signal Process. Lett.*, vol. 26, pp. 961–965, Jul. 2019.
- [7] S. Sedighi, B. S. M. R. M. Soltanalian, and B. Ottersten, "On the performance of one-bit DoA estimation via sparse linear arrays," *IEEE Trans. Signal Process.*, vol. 69, pp. 6165–6182, Oct. 2021.
- [8] Y.-H. Xiao, L. Huang, D. Ramírez, C. Qian, and H. C. So, "Covariance matrix recovery from one-bit data with non-zero quantization thresholds: Algorithm and performance analysis," *IEEE Trans. Signal Process.*, vol. 71, pp. 4060 – 4076, Nov. 2023.
- [9] L. Huang, S. Wu, and X. Li, "Reduced-rank MDL method for source enumeration in high-resolution array processing," *IEEE Trans. Signal Process.*, vol. 55, pp. 5658–5667, Dec. 2007.

- [10] L. Huang and H. C. So, "Source enumeration via MDL criterion based on linear shrinkage estimation of noise subspace covariance matrix," *IEEE Trans. Signal Process.*, vol. 61, pp. 4806–4821, Oct. 2013.
- [11] M. Wax and A. Adler, "Detection of the number of signals by signal subspace matching," *IEEE Trans. Signal Process.*, vol. 69, pp. 973–985, 2021.
- [12] X.-P. Li, Z.-L. Shi, L. Huang, A. M.-C. So, and H. C. So, "ROCS: Robust one-bit compressed sensing with application to direction of arrival," *IEEE Trans. Signal Process.*, vol. 72, pp. 2407 – 2420, Apr. 2024.
- [13] Z. Yang, J. Li, P. Stoica, and L. Xie, "Chapter 11 - sparse methods for direction-of-arrival estimation," in *Academic Press Library in Signal Processing, Volume 7* (R. Chellappa and S. Theodoridis, eds.), pp. 509–581, Academic Press, 2018.
- [14] C. Stöckle, J. Munir, A. Mezghani, and J. A. Nossek, "1-bit direction of arrival estimation based on compressed sensing," in *16th IEEE Workshop Signal Process. Adv. Wireless Commun.*, (Stockholm, Sweden), pp. 246–250, Jul. 2015.
- [15] P. Wang, H. Yang, and Z. Ye, "1-bit direction of arrival estimation via improved complex-valued binary iterative hard thresholding," *Digit. Signal Process.*, vol. 120, p. 103265, Jan. 2022.
- [16] Z. Wei, W. Wang, F. Dong, and Q. Liu, "Gridless one-bit direction-of-arrival estimation via atomic norm denoising," *IEEE Commun. Lett.*, vol. 24, pp. 2177–2181, Oct. 2020.
- [17] X. Meng and J. Zhu, "A generalized sparse bayesian learning algorithm for 1-bit DOA estimation," *IEEE Commun. Lett.*, vol. 22, pp. 1414–1417, Jul. 2018.
- [18] L. Feng, L. Huang, Q. Li, Z.-Q. He, and M. Chen, "An off-grid iterative reweighted approach to one-bit direction of arrival estimation," *IEEE Trans. Veh. Technol.*, vol. 72, pp. 8134–8139, Jan. 2023.
- [19] X.-P. Li, Z.-L. Shi, L. Huang, Y.-H. Xiao, and H. C. So, "Robust single-bit direction-of-arrival estimation with magnitude recovery," *IEEE J. Sel. Topics Signal Process.*, 2025.
- [20] M. Chen, Q. Li, L. Huang, L. Feng, M. Rihan, and D. Xia, "One-bit DOA estimation using robust sparse covariance fitting in non-uniform noise," in *Proc. 12th Sensor Array and Multichannel Signal Process. Workshop*, (Trondheim, Norway), pp. 11–15, Jun. 2022.
- [21] X. P. Li, Z.-Y. Wang, Z.-L. Shi, H. C. So, and N. D. Sidiropoulos, "Robust tensor completion via capped frobenius norm," *IEEE Trans. Neural Netw. Learn. Syst.*, vol. 35, no. 7, pp. 9700–9712, 2023.
- [22] M. Chen, Q. Li, X. P. Li, L. Huang, and M. Rihan, "One-bit DoA estimation for deterministic signals based on $\ell_{2,1}$ -norm minimization," *IEEE Trans. Aerosp. Electron. Syst.*, vol. 60, no. 2, pp. 2438–2444, 2024.
- [23] X. P. Li, Z.-L. Shi, Q. Liu, and H. C. So, "Fast robust matrix completion via entry-wise ℓ_0 -norm minimization," *IEEE Trans. Cybern.*, vol. 53, pp. 7199–7212, Nov. 2023.
- [24] X. P. Li, Z. Liu, Z.-L. Shi, and H. C. So, "MUSIC with capped frobenius norm: Efficient robust direction-of-arrival estimator," *IEEE Trans. Aerosp. Electron. Syst.*, vol. 59, pp. 8090–8103, Dec. 2023.
- [25] Z. Liu, X. P. Li, and H. C. So, " ℓ_0 -norm minimization-based robust matrix completion approach for MIMO radar target localization," *IEEE Trans. Aerosp. Electron. Syst.*, vol. 59, pp. 6759–6770, Oct. 2023.
- [26] M. Chen, Q. Li, L. Huang, L. Feng, and M. Rihan, "One-bit Cramér–Rao Bound of direction of arrival estimation for deterministic signals," *IEEE Trans. Circuits Syst. II Exp. Briefs*, Feb. 2024.